

Heaven's Light is Our Guide

Rajshahi University of Engineering & Technology



Department of Mechatronics Engineering

Course No.: MTE 3106

Course Title: Linear Integrated Circuits and Digital Systems Sessional

Experiment No.: 05

Experiment Name: Experiment Name: Design and Implementation of SISO, SIPO, PISO, PIPO, Johnson, and Ring Counters in Logisim

Remarks

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Date of Experiment:

Date of Submission:

Experiment No: 05

Experiment Name: Design and Implementation of SISO, SIPO, PISO, PIPO, Johnson, Ring counters in Logisim.

Objectives:

- Understand the fundamental operations of different types of shift registers (SISO, SIPO, PISO, and PIPO) and counters (Johnson, Ring), as well as their roles in sequential digital circuits.
- Design logic circuits for each type of shift register and counter using flip-flops and the required combinational logic components.
- Implement and simulate the designed circuits using Logisim software to verify their theoretical functionality.

Theory:

A register is a group of flip-flops, each sharing a common clock and capable of storing one bit of information. An n -bit register consists of a group of n flip-flops capable of storing n bits of binary information. In addition to the flip-flops, a register may feature combinational gates that perform specific data-processing tasks. In its broadest definition, a register consists of a group of flip-flops combined with gates that dictate how information is transferred into the register.

A register capable of shifting its stored binary information to its neighboring cell in a selected direction is called a shift register. The logical configuration of a shift register consists of a chain of cascaded flip-flops, with the output of one flip-flop connected to the input of the next. All flip-flops receive common clock pulses, which activate the shift of data from one stage to the next. [1]

SISO Shift Register:

The SISO shift register is the most basic form of data shifting circuit. It accepts binary data serially (one bit at a time) through a single input pin and shifts it out serially through a single output pin. Internally, the output of each flip-flop connects directly to the input of the successive one.

Because data must pass through every single flip-flop in the chain, it requires n clock pulses to fully load an n -bit word into the register, and another n clock pulses to shift it out completely. While this makes the SISO register relatively slow for data transfer, it is incredibly useful as a digital delay line. By adjusting the clock frequency or the number of flip-flops in the cascade, engineers can introduce a precise time delay into a digital signal, allowing it to act as a synchronized buffer between asynchronous devices. [2]

CLK	Q ₃	Q ₂	Q ₁	Q ₀
Initially	0	0	0	0
0	1	0	0	0
0	1	1	0	0
0	1	1	1	0
0	1	1	1	1

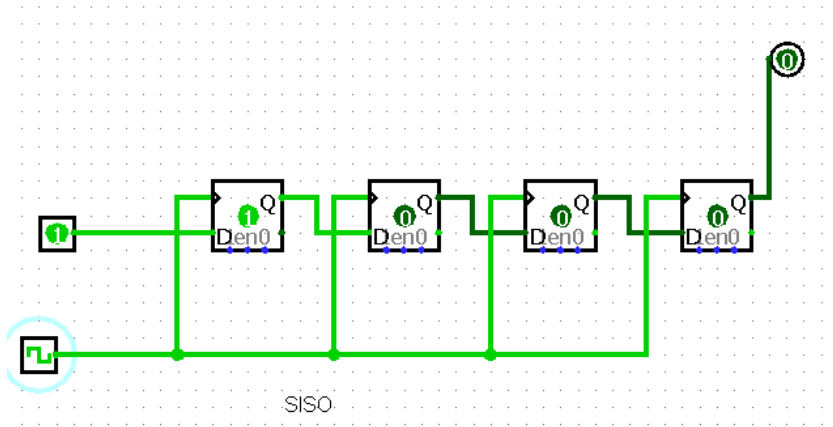


Figure 1: SISO Shift Register Logisim circuit diagram

SIPO Shift Register:

The SIPO shift register is primarily used as a serial-to-parallel converter. Like the SISO register, data is fed into the system serially, bit by bit, requiring n clock cycles to load an n -bit word. However, the key structural difference is that the outputs of *every* flip-flop are externally accessible simultaneously. Once the entire data string has been clocked into the register, the complete binary word can be read across the parallel output lines all at once. This configuration is widely used in communication systems (such as UART receivers) where data is transmitted over a single wire to save space, but needs to be processed by a microprocessor in parallel chunks (like 8-bit bytes) for faster internal computation. [2]

CLK	Q ₃	Q ₂	Q ₁	Q ₀
Initially	0	0	0	0
0	1	0	0	0
0	1	1	0	0
0	1	1	1	0
0	1	1	1	1

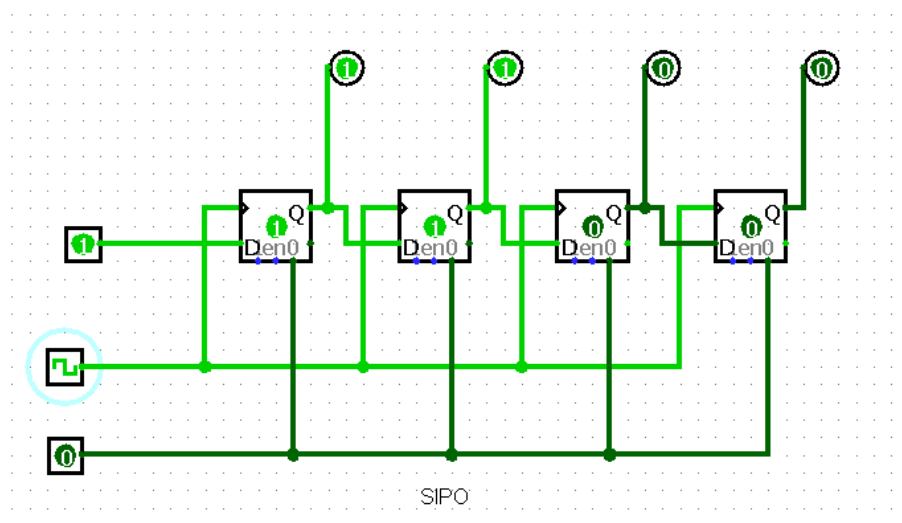


Figure 2: SIPO Shift Register Logisim circuit diagram

PISO Shift Register:

The PISO shift register functions as a parallel-to-serial converter, performing the exact opposite role of the SIPO circuit. It takes parallel data from multiple input lines and transmits it out sequentially over a single output line. To achieve this, the PISO register utilizes a control signal typically called "Shift/Load". When this control line is set to the "Load" state, combinational logic (such as multiplexers) bypasses the shifting mechanism, allowing all n bits of input data to be jammed into the flip-flops simultaneously on a single clock pulse. Once the data is loaded, the control line is switched to "Shift", and subsequent clock pulses push the stored data out of the final flip-flop sequentially. This is a critical component in digital transmitters, taking parallel data from a computer bus and formatting it for transmission over a single-wire network. [2]

CLK	Load	Q ₃	Q ₂	Q ₁	Q ₀	Output
0	0	0	1	0	1	0
1	1	0	1	0	1	0
1	0	1	1	1	1	0
1	1	1	1	1	0	1
1	1	1	1	0	1	1
1	1	1	0	1	1	1

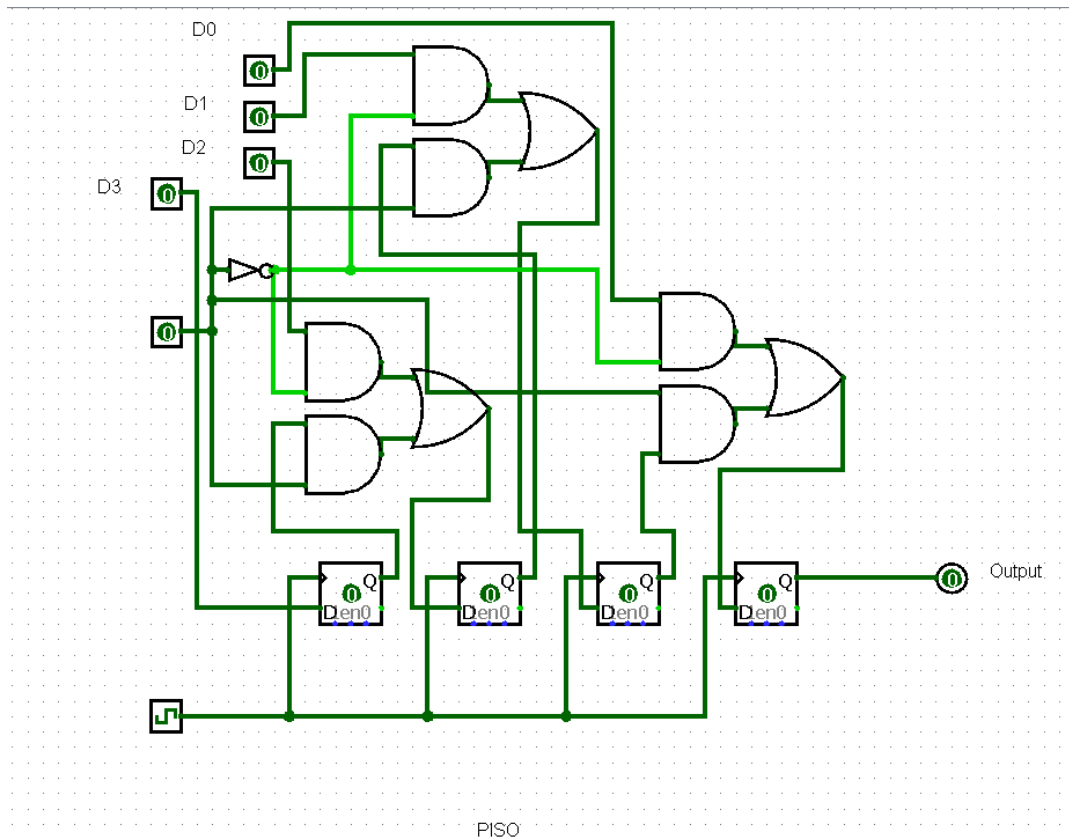


Figure 3: PISO Shift Register Logisim circuit diagram

PIPO Shift Register:

The PIPO register is the fastest of the four types because it handles all data loading and outputting simultaneously. It features dedicated parallel input lines (PI) and parallel output lines (PO). In a PIPO circuit, there is no physical interconnection between the output of one flip-flop and the input of the next; therefore, no actual "shifting" of data occurs between the internal cells. Instead, upon the arrival of the clock edge, the data present on the input lines is instantly transferred to the output lines in a single clock cycle. This configuration acts as a multi-bit data latch, serving as temporary memory storage or a synchronization buffer holding data steady while other parts of a microprocessor complete their operations. [2]

CLK	Load	Q3	Q2	Q1	Q0	Output
0	0	0	0	0	0	0000
1	1	1	0	1	0	1010
1	1	0	1	0	1	0101
1	1	1	1	0	0	1100
1	1	0	0	1	1	0011
1	1	1	1	1	1	1111

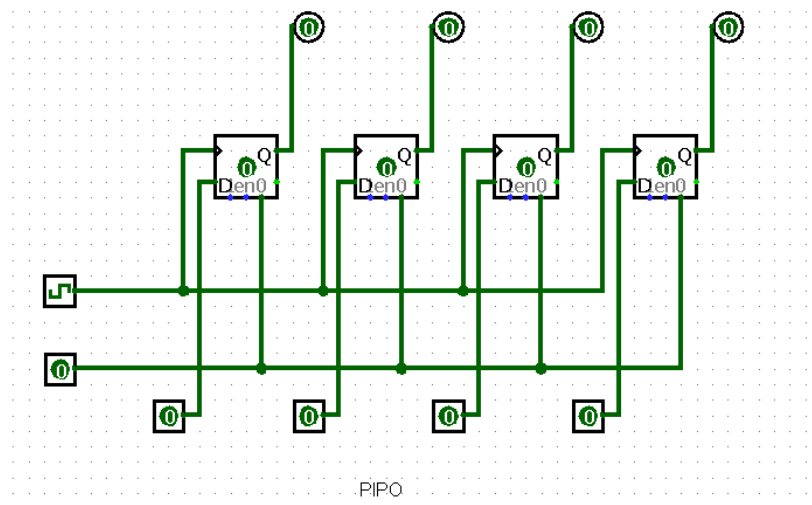


Figure 4: PIPO Shift Register Logisim circuit diagram

Johnson Counter

A Johnson Counter (also known as a Twisted Ring Counter) is an advanced variation of a shift register where the inverted output ($\bar{Q}$ or complemented output) of the final flip-flop is fed back into the data input of the first flip-flop. Because of this twisted feedback loop, an n -stage Johnson counter cycles through exactly $2n$ distinct states before repeating. For example, a 4-bit Johnson counter will produce 8 unique states. The sequence fills with 1s from left to right, and then empties with 0s from left to right. This behavior requires significantly less decoding logic than a traditional binary counter. As a result, Johnson counters are highly favored in hardware design for creating complex timing sequences, multi-phase clock generators, and glitch-free digital frequency dividers.

CLK	Q3	Q2	Q1	Q0	Output
0	0	0	0	0	0000
1	1	0	0	0	1000
1	1	1	0	0	1100
1	1	1	1	0	1110
1	1	1	1	1	1111
1	0	1	1	1	0111
1	0	0	1	1	0011
1	0	0	0	1	0001

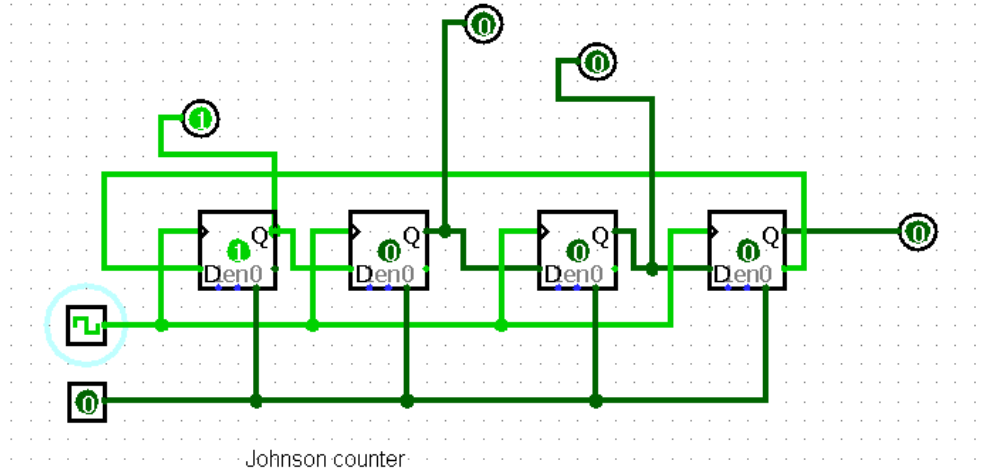


Figure 5: Johnson Counter Logisim circuit diagram

Ring Counter:

A Ring Counter is a straightforward circular shift register where the standard, non-inverted output (Q) of the last flip-flop is looped directly back to the input of the first flip-flop. Unlike the Johnson counter, a Ring counter typically requires a pre-set or initialization step where a single logic '1' is injected into the first flip-flop while the rest are set to '0'. Once the clock starts, this single '1' simply rotates endlessly through the chain of flip-flops. Because only one output is high at any given moment, an n -bit ring counter provides n distinct states. This "one-hot" state encoding makes the Ring Counter exceptionally easy to decode, eliminating the need for external logic gates when stepping through a specific sequence of operations in a microcontroller or state machine. [2]

CLK	Q3	Q2	Q1	Q0	Output
0	1	0	0	0	1000
1	0	1	0	0	0100
1	0	0	1	0	0010
1	0	0	0	1	0001
1	1	0	0	0	1000
1	0	1	0	0	0100

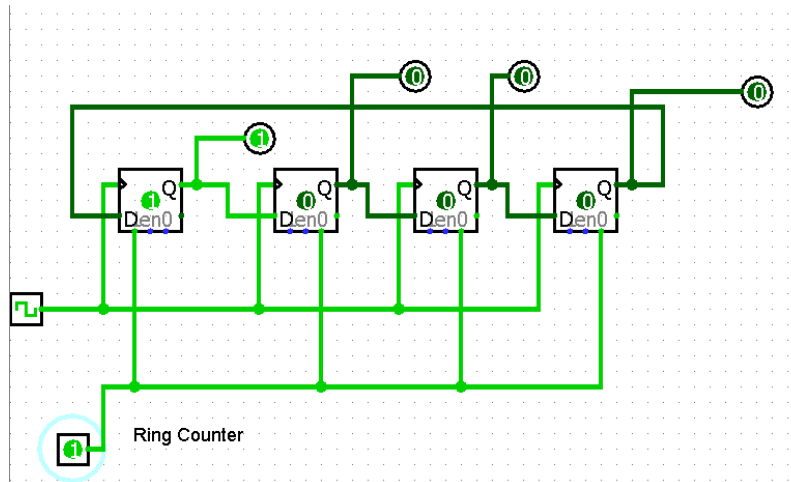


Figure 6: Ring Counter Logisim circuit diagram

Discussion:

In this experiment, different types of shift registers including SISO, SIPO, PISO, and PIPO—along with Ring Counter and Johnson Counter circuits, were designed and simulated using Logisim. The shift registers demonstrated various methods of data storage and transfer through serial and parallel input-output configurations controlled by clock pulses. The SISO and SIPO registers successfully illustrated the conversion of serial data into serial and parallel outputs, respectively. Conversely, the PISO and PIPO registers demonstrated the handling of parallel data with serial or parallel outputs. Additionally, the Ring Counter exhibited the circulation of a single logic state through a sequence of flip-flops, whereas the Johnson Counter generated a unique counting sequence by feeding the inverted output of the last flip-flop back to the first. The simulation results for all circuits were consistent with their theoretical operations, firmly verifying the correctness of the designs.

Conclusion:

This experiment provided valuable insight into the operation and practical applications of shift registers and counters in digital electronics. Through Logisim simulation, the principles of data storage, data shifting, and sequence generation were successfully demonstrated. The behavior of the SISO, SIPO, PISO, and PIPO shift registers, as well as the Ring and Johnson Counters, perfectly matched the expected theoretical outcomes. This successfully fulfills the objectives of the experiment and strengthens our foundational understanding of sequential logic systems.

References:

- [1] M. Morris Mano and Michael D. Ciletti, Digital Design, 5th ed. Upper Saddle River, NJ: Pearson Education, 2013.
- [2] SISO Shift Register: Circuit, Working, Waveforms & Its Applications,” *ElProCus*, published approximately 3 years ago. [Online]. Available: <https://www.elprocus.com/iso-shift-register/>.